Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The slope of the line touching both the parabolas $y^2 = 4x$ and $x^2 = -32y$ is

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) $\frac{1}{8}$

(D) $\frac{2}{3}$

Q.2

If the tangents at the extremities of a chord PQ of a parabola intersect at T, then the distances of the focus of the parabola from the points P, T, Q are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

Q.3

The orthocentre of the triangle formed by three distinct tangents to the parabola $y^2 = 4ax$ lies on:

(A) $x + a = 0$

(B) $x - a = 0$

(C) $x = 0$

(D) $y = 0$

Q.4

If the focus of a parabola divides a focal chord of the parabola in segments of length 3 and 2, the length of the latus rectum of the parabola is

(A) $3/2$

(B) $6/5$

(C) $12/5$

(D) $24/5$

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Q.5

If PQ is a focal chord of the parabola $y^2 = 4ax$ with focus at S, then $\frac{2\,SP.SQ}{SP+SQ} =$

(A) a

(B) $2a$

(C) $4a$

(D) a^2

Q.6

Tangents at the extremities of a focal chord of a parabola intersect

(A) on the axis of the parabola

(B) on the tangent at the vertex

(C) at the point of intersection of the directrix and the line parallel to the axis of the parabola through the mid-point of the chord.

(D) none of these

Q.7

If the area of the triangle inscribed in the parabola $y^2 = 4ax$ with one vertex at the vertex of the parabola and other two vertices at the extremities of a focal chord is $5a^2/2$, then the length of the focal chord is

(A) $3a$

(B) $5a$

(C) $25a/4$

(D) none of these

Q.8

A triangle ABC is inscribed in the parabola $y^2 = 4x$ such that A lies at the vertex of the parabola and BC is a focal chord of the parabola with one extremity at $(9, 6)$, the centroid of the triangle ABC lies at

(A) $(41/27, 8/9)$

(B) $(82/27, 16/9)$

(C) $(87/27, 20/9)$

(D) $(80/27, 20/9)$

Q.9

Locus of the mid points of the chords of the parabola $y^2 = 8x$ which touch the circle $x^2 + y^2 = 4$ is

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(A) $\left(y^2-4x\right)^2=16\left(x^2+4\right)$

(B) $\left(y^2-4x^2\right)=4\left(x^2+a^2\right)$

(C) $\left(y^2-4x\right)^2=4\left(16+y^2\right)$

(D) $\left(y^2-4x\right)^2=16\left(4+y^2\right)$

Q.10

Through the vertex of the parabola $y^2=4ax$, chords OA and OB are drawn at right angles to each other. For all positions of the point A, the chord AB meets the axis of the parabola at a fixed point. Coordinates of the fixed point are:

(A) $(a,0)$

(B) $(-a,0)$

(C) $(4a,0)$

(D) $(-4a,0)$

Q.11

If the tangents at P and Q on a parabola meet in T, then SP, ST and SQ are in

(A) A.P.

(B) GP.

(C) H.P.

(D) None of these

Q.12

The equation of the parabola whose focus is $(0,0)$ and the tangent at the vertex is $\mathrm{x}-\mathrm{y}+1=0$ is

(A) $x^2+y^2+2xy-4x+4y-4=0$

(B) $x^2-4x+4y-4=0$

(C) $y^2-4x+4y-4=0$

(D) $2x^2+2y^2-4xy-x+y-4=0$

Q.13

If three points $\mathrm{E}, \mathrm{F}, \mathrm{G}$ are taken on the parabola $\mathrm{y}^2=4\mathrm{axso}$ that their ordinates are in G.P., then the tangents at E and G intersect on the

(A) directrix

(B) axis

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(C) ordinate of F

(D) tangent at F

Q.14

The equation of the parabola whose focus is $(0,0)$ and the tangent at the vertex is $x-y+1=0$ is

(A) $x^2+y^2+2xy-4x+4y-4=0$

(B) $x^2-4x+4y-4=0$

(C) $y^2-4x+4y-4=0$

(D) $2x^2+2y^2-4xy-x+y-4=0$

Q.15

The equation of a tangent to the parabola $y^2=8x$ is $y=x+2$. The point on this line from which the other tangent to the parabola is perpendicular to the given tangent is

(A) $(2,4)$

(B) $(-2,0)$

(C) $(-1,1)$

(D) $(0,2)$

Q.16

The equation of the parabola whose focus is the point $(0,0)$ and the tangent at the vertex is $x-y+1=0$, is

(A) $x^2+y^2-2xy-4x+4y-4=0$

(B) $x^2+y^2-2xy+4x-4y-4=0$

(C) $x^2+y^2+2xy-4x+4y-4=0$

(D) $x^2+y^2+2xy-4x-4y+4=0$

Q.17

Through the vertex O of a parabola $\mathrm{y^2=4x}$, chords OP and OQ are drawn at right angles to one another. The locus of the middle point of PQ is

(A) $y^2=2x+8$

(B) $y^2=x+8$

(C) $y^2=2x-8$

(D) $y^2=x-8$

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Q.18

If y_1, y_2 are the ordinates of two points P and Q on the parabola and y is the ordinate of the point of intersection of tangents at P and Q, then

(A) y_1, y_2, y_3 are in A.P.

(B) y_1, y_3, y_2 are in A.P.

(C) y_1, y_2, y_3 are in G.P.

(D) y_1, y_3, y_2 are in G.P.

Q.19

Equation of the tangent to the parabola $x^2 = 4y$ perpendicular to the line $y = 3x + 2$ is

(A) $3x + 9y - 1 = 0$

(B) $x + 3y - 1 = 0$

(C) $x + 3y + 1 = 0$

(D) $3x + 9y + 1 = 0$

Q.20

What is the area of the triangle formed by the lines joining the vertex of the parabola $x^2 = 12y$ to the ends of the latus rectum?

(A) 9 square units

(B) 12 square units

(C) 14 square units

(D) 18 square units

Answers & Solutions

Q.1 Answer:

$\frac{1}{2}$

Solution:

Equation of the tangent at $\left(t^2, 2t\right)$ **to the parabola** $y^2 = 4x$ **is** $ty = x + t^2$ **(1)**
and the tangent at $\left(-16t', -8t'^2\right)$ **to the parabola** $x^2 = -32y$ **is**
$t'x = y - 8t'^2$ **(2)**
Since (1) and (2) represent the same line, comparing we get
$\frac{1}{-t'} = \frac{-t}{1} = \frac{t^2}{-8t'^2} \Rightarrow t = 2$
Hence the required slope of the tangent is $\frac{1}{t} = \frac{1}{2}$**.**

Q.2 Answer:

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G.P.

Solution:

Let the equation of the parabola be $y^2 = 4ax$ and $P\left(at_1^2, 2at_1\right), Q\left(at_2^2, 2at_2\right)$ be the extremities of the chord PQ.

The coordinates of T, the point of intersection of the tangents at P and Q are

$$\left(at_1t_2, a\left(t_1 + t_2\right)\right)$$

Now $SP = a\left(1 + t_1^2\right)$

$SQ = a\left(1 + t_2^2\right)$

and $ST^2 = \left(at_1t_2 - a\right)^2 + \left[a\left(t_1 + t_2\right) - 0\right]^2$

$= a^2\left(t_1^2 + t_2^2 + t_1^2t_2^2 + 1\right)$

$= a^2\left(1 + t_1^2\right)\left(1 + t_2^2\right) = SP.SQ$

So that SP, ST, SQ are in G.P.

Q.3 Answer:

$x + a = 0$

Solution:

Equations of the tangents at the points $A\left(at_1^2, 2at_1\right), B\left(at_2^2, 2at_2\right)$ and $C\left(at_3^2, 2at_3\right)$ are

$t_1y = x + at_1^2$

$t_2y = x + at_2^2$

$t_3y = x + at_3^2$

These tangents meet at

$P\left(at_2t_3, a\left(t_3 + t_2\right)\right)$

$Q\left(at_3t_1, a\left(t_3 + t_1\right)\right)$

$R\left(at_1t_2, a\left(t_1 + t_2\right)\right)$

Equation of the altitude through P is $y - a\left(t_2 + t_3\right) = -t_1\left(x - at_2t_3\right)$

$\Rightarrow y - a\left(t_1 + t_2 + t_3 + t_1t_2t_3\right) = -t_1(x + a)$

which passes through the point $\left(-a, a\left(t_1 + t_2 + t_3 + t_1t_2t_3\right)\right)$

From symmetry the other two altitudes also pass through this point and hence the orthocentre of the triangle lies on $x = -a$ or $x + a = 0$.

Q.4 Answer:

$24/5$

Solution:

Let $y^2 = 4ax$ be the equation of the parabola, then the focus is $S(a, 0)$. Let $P\left(at^2{}_1, 2at_1\right)$ and $Q\left(at_2^2, 2at_2\right)$ be vertices of a focal chord of the parabola, then $t_1t_2 = -1$. Let $SP = 3, SQ = 2$

$SP = \sqrt{a^2\left(1 - t_1^2\right)^1 + 4a^2t_1^2} = a\left(1 + t_1^2\right) = 3$ **(i)**

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and $SQ = a\left(1 + \frac{1}{t_1^2}\right) = 2$ **(ii)**

From (i) and (ii) we get $t_1^2 = 3/2$ **and** $a = 6/5$ **Hence the length of the latus rectum** $= 24/5$**.**

Q.5 Answer:

$2a$

Solution:

Let the coordinates of P **be** $\left(at_1^2, 2at_1\right)$ **and of** Q **be** $\left(at_2^2, 2at_2\right)$ **.**

Since PQ **is a focal chord,** $t_1 t_2 = -1$

Focus is $S(a,0) \Rightarrow SP = \sqrt{a^2\left(1 - t_1^2\right)^2 + 4a^2 t_1^2} = a\left(1 + t_1^2\right)$

and $SQ = a\left(1 + 1/t_1^2\right) = \frac{a\left(1+t_1^2\right)}{t_1^2}$

So that $\frac{2SP \cdot SQ}{SP+SQ} = \frac{2a^2\left(1+t_1^2\right)^2}{t_1^2 a\left[\left(1+t_1^2\right)+\left(1+\frac{1}{t_1^2}\right)\right]} = 2a$

Q.6 Answer:

at the point of intersection of the directrix and the line parallel to the axis of the parabola through the mid-point of the chord.

Solution:

Extremities of a focal chord are $\left(at^2, 2at\right)$ **,** $\left(a/t^2, -2a/t\right)$

Tangents are $ty = x + at^2$**, and** $-ty = xt^2 + a$**.**

They intersect at $\left(-a, \frac{a\left(t^2-1\right)}{t}\right)$ **or** $\left(-a, a\left(t - \frac{1}{t}\right)\right)$ **which is the point of intersection of the directrix** $x = -a$ **and the line** $y = a\left(t - \frac{1}{t}\right)$ **which is the line through the mid-point of the chord parallel to the axis.**

Q.7 Answer:

$25a/4$

Solution:

Let the vertices be $O(0,0), A\left(at^2, 2at\right),$

$B\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$ **then**

$\frac{1}{2}\left|\begin{vmatrix} 0 & 0 & 1 \\ at^2 & 2at & 1 \\ \frac{a}{t^2} & \frac{2a}{t} & 1 \end{vmatrix}\right| = \frac{5a^2}{2} \Rightarrow 2t^2 - 5t + 2 = 0$

$\Rightarrow t = 2$ **or** $1/2$ **so the vertices of a focal chord are** $(4a, 4a)$ **and** $(a/4, -a)$ **(Taking** $t = 2$**) and length of this focal chord is** $25\mathrm{a}/4$**.**

Q.8 Answer:

$(82/27, 16/9)$

Solution:

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If one extremity of the focal chord is $(9, 6)$, the other is $(1/9, -2/3)$ so the centroid of the triangle is $\left(\frac{0+9+1/9}{3}, \frac{0+6-2/3}{3}\right) = \left(\frac{82}{27}, \frac{16}{9}\right)$

Q.9 Answer:

$\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

Solution:

Let (h, k) be the mid-point of the chord of the parabola $y^2 = 8x$, then its equation is

$ky - 4(x + h) = k^2 - 8h$

$\Rightarrow 4x - ky + k^2 - 4h = 0$

which touches the circle $x^2 + y^2 = 4$

$\Rightarrow \frac{k^2 - 4h}{\sqrt{16 + k^2}} = 2$

$\Rightarrow \left(k^2 - 4h\right)^2 = 4\left(k^2 + 16\right)$

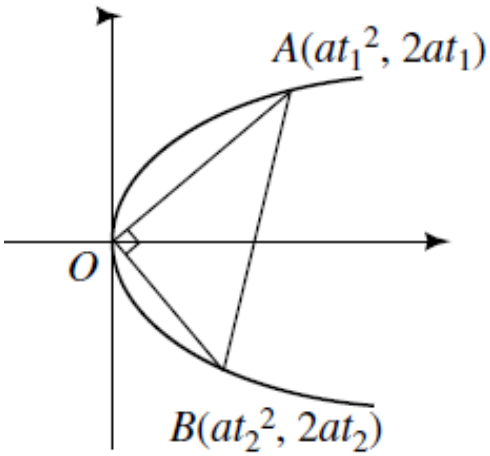
Locus of (h, k) is $\left(y^2 - 4x\right)^2 = 4\left(16 + y^2\right)$

Q.10 Answer:

$(4a, 0)$

Solution:

Let the coordinates of A be $\left(at_1^2, 2at_1\right)$ and of B be $\left(at_2^2, 2at_2\right)$



OA is perpendicular to OB

$\Rightarrow \left(\frac{2}{t_1}\right)\left(\frac{2}{t_2}\right) = -1 \Rightarrow t_1 t_2 = -4$

Equation of AB is

$\frac{x - at_1^2}{at_2^2 - at_1^2} = \frac{y - 2at_1}{2at_2 - 2at_1}$

$\Rightarrow 2\left(x - at_1^2\right) = \left(t_1 + t_2\right)\left(y - 2at_1\right)$

$\Rightarrow 2\left(x + at_1 t_2\right) - y\left(t_1 + t_2\right) = 0$

$\Rightarrow 2(x - 4a) - y\left(t_1 + t_2\right) = 0$

which meets the axis $y = 0$ at the point $(4a, 0)$ for all values of t_1.

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Q.11 Answer:

GP.

Solution:

Let $\mathrm{P}\left(\mathrm{at}_1^2, 2\mathrm{at}_1\right)$ **and** $\mathrm{Q}\left(\mathrm{at}_2^2, 2\mathrm{at}_2\right)$ **be two points on the parabola** $y^2 = 4ax$**. The tangents at P and Q intersect at** $\mathrm{T}\left[\mathrm{at}_1\, \mathrm{t}_2, \mathrm{a}\left(\mathrm{t}_1 + \mathrm{t}_2\right)\right]$**.**

Now, $SP = \sqrt{\left(at_1^2 - a\right)^2 + 2\left(at_1 - 0\right)^2}$

$= a\left(t_1^2 + 1\right)$

SQ $= a\left(t_2^2 + 1\right)$

and $ST = \sqrt{\left(at_1t_2 - a\right)^2 + \left[a\left(t_1 + t_2\right) - 0\right]^2}$

$= a\sqrt{\left(1 + t_1^2\right)\left(1 + t_2^2\right)}$

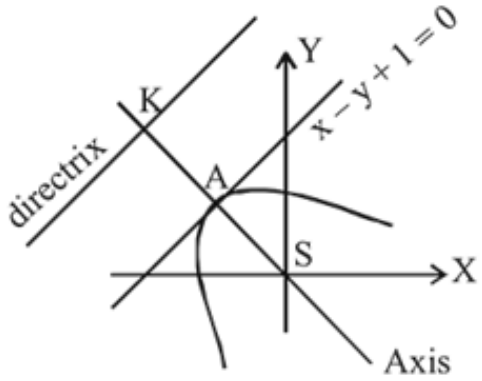
$\Rightarrow ST^2 = a^2\left(1 + t_1^2\right)\left(1 + t_2^2\right) =$ **SP.SQ**

Hence, SP, ST, SQ are in G.P.

Q.12 Answer:

$x^2 + y^2 + 2xy - 4x + 4y - 4 = 0$

Solution:



The length of the perpendicular drawn from the given focus upon the given line $x - y + 1 = 0$ **is** $\frac{0-0+1}{\sqrt{(1)^2+(-1)^2}} = \frac{1}{\sqrt{2}}$

The directrix is parallel to the tangent at the vertex. So, the equation of the directrix is $\mathrm{x} - \mathrm{y} + \lambda = 0$**, where** λ **is a constant to be determine. But the distance between the focus and the directrix** $= 2 \times$ **(the distance between the focus and the tangent at the vertex)**

$= 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$**. Hence** $\frac{0-0+\lambda}{\sqrt{(1)^2+(-1)^2}} = \sqrt{2}$**.**

$\therefore \lambda = 2.[\lambda$ **must be positive see figure** $] \therefore$ **The directrix is the line** $x - y + 2 = 0$**. Let** (x, y) **be a moving point on the parabola. By the focus-directrix property of the parabola, its equation is**

$(x - 0)^2 + (y - 0)^2 = \left(\pm\frac{x-y+2}{\sqrt{2}}\right)^2$

or $x^2 + y^2 + 2xy - 4x + 4y - 4 = 0$

Q.13 Answer:

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ordinate of F

Solution:

Given parabola is $y^2 = 4ax$ ...(1) Let the coordinates of E, F and G be respectively $\left(at_1^2, 2at_1\right)$, $\left(at_2^2, 2at_2\right)$ and $\left(at_3^2, 2at_3\right)$

Since ordinates of E, F and G are in G.P.

$\therefore \left(2\mathrm{at}_2\right)^2 = \left(2\mathrm{at}_1\right)\left(2\mathrm{at}_3\right)$ **or** $\mathrm{t}_2^2 = \mathrm{t}_1\mathrm{t}_3$

The tangents at E and G are $\mathrm{t}_1\mathrm{y} = \mathrm{x} + \mathrm{at}_1^2$

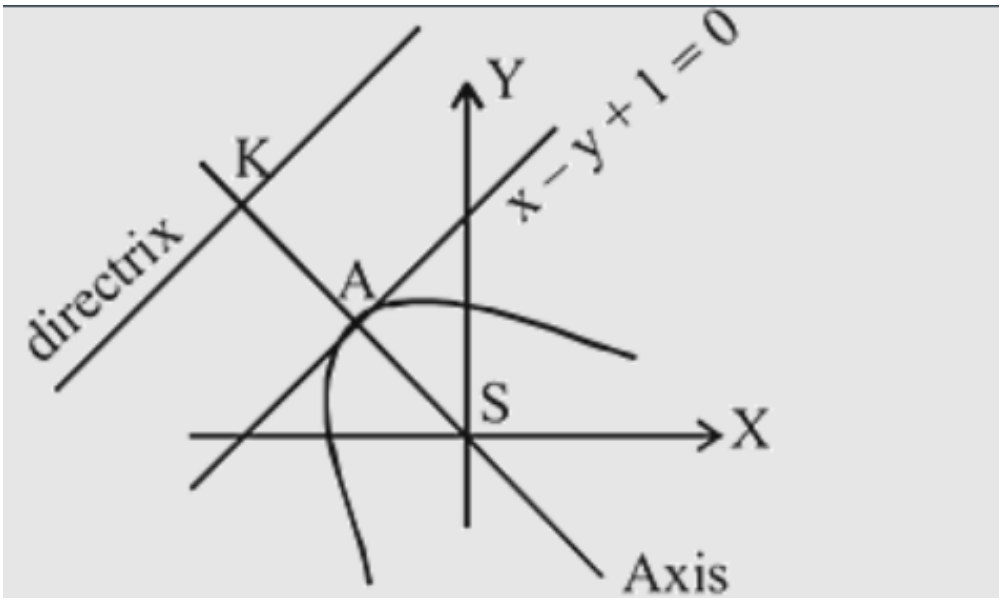
and $\mathrm{t}_3\mathrm{y} = \mathrm{x} + \mathrm{at}_3^2$

Solving (3) and (4), we get $x = at_1t_3 = at_2^2$ $[$ from $(2)]$ Since the x -coordinate of the point of intersection is at 2_2, the point lies on the line $x = at_2^2$ i.e. on the ordinate of F $\left(\mathrm{at}_2^2, 2\mathrm{at}_2\right)$

Q.14 Answer:

$x^2 + y^2 + 2xy - 4x + 4y - 4 = 0$

Solution:



The length of the perpendicular drawn from the given focus upon the given line $x - y + 1 = 0$ is $\frac{0-0+1}{\sqrt{(1)^2+(-1)^2}} = \frac{1}{\sqrt{2}}$.

The directrix is parallel to the tangent at the vertex. So, the equation of the directrix is $x - y + \lambda = 0$, where λ is a constant to be determine. But the distance between the focus and the directrix $= 2\times$ (the distance between the focus and the tangent at the vertex) $= 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$

Hence $\frac{0-0+\lambda}{\sqrt{(1)^2+(-1)^2}} = \sqrt{2}.$

$\therefore \lambda = 2.$

$[\lambda$ **must be positive see figure** $]\therefore$ **The directrix is the line** $\mathrm{x} - \mathrm{y} + 2 = 0$**. Let** (x, y) **be a moving point on the parabola. By the focusdirectrix property of the parabola, its equation is**

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$(x-0)^2+(y-0)^2=\left(\pm\frac{x-y+2}{\sqrt{2}}\right)^2$

or $x^2+y^2+2xy-4x+4y-4=0$

Q.15 Answer:

$(-2,0)$

Solution:

$y=mx+\frac{a}{m}$ **is a tangent for all values of** m **where** $a=2$.

For $m=1, y=x+2$**, so the equation of the tangent perpendicular to the given tangent is** $y=-x-2$

The required point is the point of intersection to there tangents i.e., $(-2,0)$.

Alternately perpendicular tangents to a parabola intersect on the directrix, $x+2=0$.

Q.16 Answer:

$x^2+y^2+2xy-4x+4y-4=0$

Solution:

Equation of the tangent at the vertex is

$x-y+1=0$ **.... (i)**

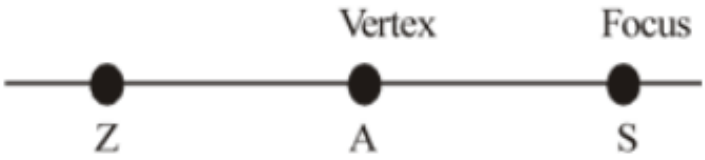
Equation of the axis of the parabola is

$x+y+k=0$ **... (ii)**

Since it passes through $(0,0)$ $0+0+k=0$

$\Rightarrow k=0$

$\therefore$ **Equation of axis is** $x+y=0$ **... (iii)**



Where z **is a point on directrix**

Solving (i) and (iii) we get $A\left(\frac{-1}{2},\frac{1}{2}\right)$

$\therefore z$ **is** $(-1,1)$

Now directrix is $x-y+c=0$ **... (iv)**

But equation (iv) is passes through z

$-1-1+c-0\Rightarrow c=2$

So directrix is $x-y+2=0\ldots(v)$

Using $\mathbf{PS=PM}$

$P^2=PM^2$

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Q.17 Answer:

$y^2 = 2x - 8$

Solution:

Given parabola is $y^2 = 4x$
Let $\mathrm{P} \equiv \left(\mathrm{t}_1^2, 2\mathrm{t}_1\right)$ **and** $\mathrm{Q} \equiv \left(\mathrm{t}_2^2, 2\mathrm{t}_2\right)$
Slope of $\mathrm{OP} = \frac{2\mathrm{t}_1}{\mathrm{t}_1^2} = \frac{2}{\mathrm{t}_1}$ **and slope of** $\mathrm{OQ} = \frac{2}{\mathrm{t}_2}$**...(1)**
Since $\mathrm{OP} \perp \mathrm{OQ}$
$\therefore \frac{4}{t_1 t_2} = -1$ **or** $t_1 t_2 = -4$**...(2)**
Let $\mathrm{R(h, k)}$ **be the middle point of** PQ**, then** $h = \frac{t_1^2 + t_2^2}{2} \quad \ldots(3)$ **and** $k = t_1 + t_2 \quad \ldots(4)$
From $(4), \mathrm{k}^2 = \mathrm{t}_1^2 + \mathrm{t}_2^2 + 2\mathrm{t}_1\mathrm{t}_2 = 2\,\mathrm{h} - 8[\mathrm{From}(2)$ **and** $(3)]$ **Hence locus of** $R(h, k)$ **is** $y^2 = 2x - 8$

Q.18 Answer:

y_1, y_3, y_2 are in A.P.

Solution:

Let the coordinates of P **and** Q**be** $\left(\alpha t_1^2, 2at_1\right)$ **and** $\left(at_2{}^2, 2at_2\right)$ **respectively. Then,** $y_1 = 2at_1$ **and** y_2
$2at_2$**. The coordinates of the point of intersection of the tangents at** P **and** Q **are** $(at_1t_2, a(t_1 + t_2))$.
$\therefore y_3 = a(t_1 + t_2) \Rightarrow y_3 = \frac{y_1 + y_2}{2}$
$\Rightarrow y_1, y_3, y_2$ **are in A.P.**

Q.19 Answer:

$3x + 9y + 1 = 0$

Solution:

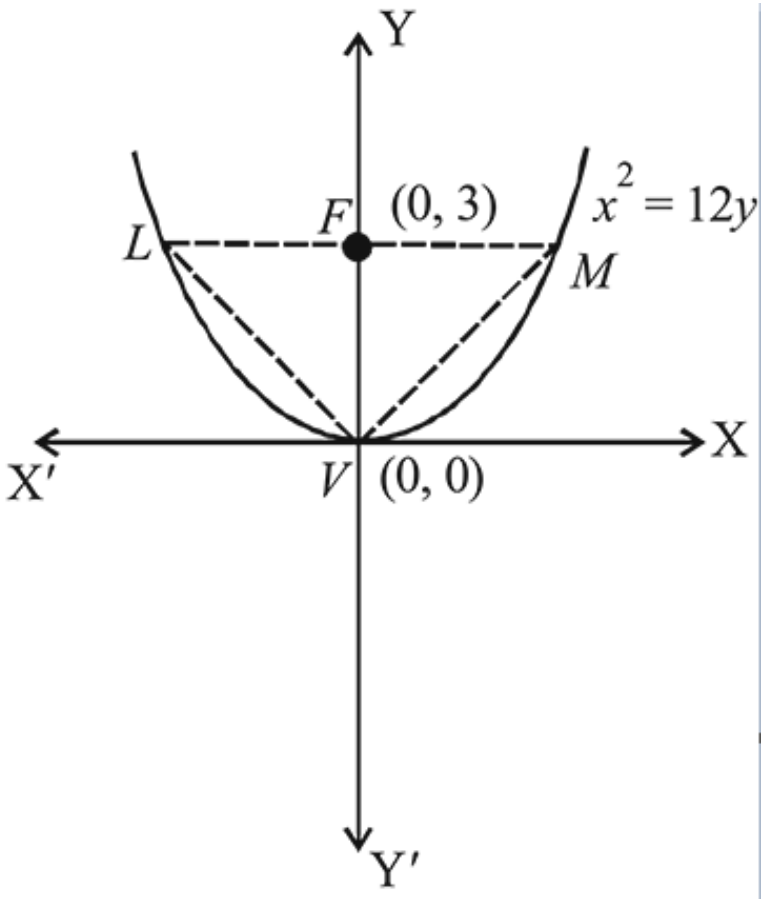
Since tangent is perpendicular to the line $y = 3x + 2$
$\therefore$ **Slope of the tangent is** $-\frac{1}{3}$
$\left(-\frac{1}{3}\right)^2 = -\frac{1}{9}$
$\therefore$ **Equation of the required tangent is**
$y = -\frac{1}{3}x - \frac{1}{9} \Rightarrow 3x + 9y + 1 = 0$

Q.20 Answer:

18 square units

Solution:

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Given parabola is $x^2 = 12y$ **which is of the form** $x^2 = 4ay$

$\Rightarrow 4a = 12 \Rightarrow a = 3$

Now, LM **is the latus rectum whose length** $= 4a = 4 \times 3 = 12$

So, area of $\Delta LMV = \frac{1}{2} \times LM \times VF.$

$= \left(\frac{1}{2} \times 12 \times 3\right)$ **sq. unit** $= 18$ **square unit**

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